

UG3 • FIRST THEORY LECTURE

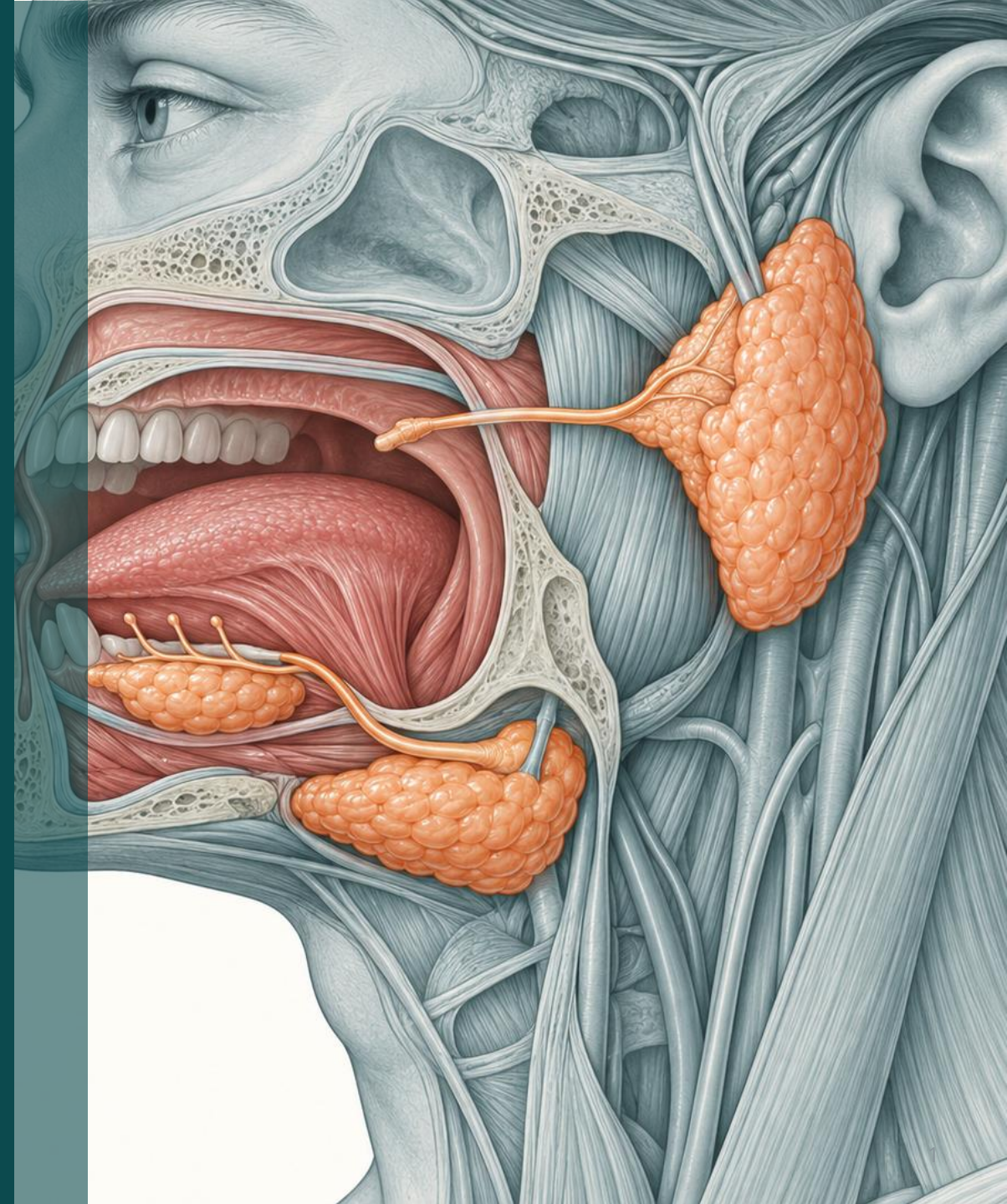
GIT Physiology LG 1

Physiology of Saliva & Swallowing

From preparing the bolus to protecting the airway and delivering food to the esophagus

Hawler Medical University • College of Medicine
Medical Physiology • AY 2026–2027

Assist. Prof. Dr. Sherwan Rahman Sulaiman

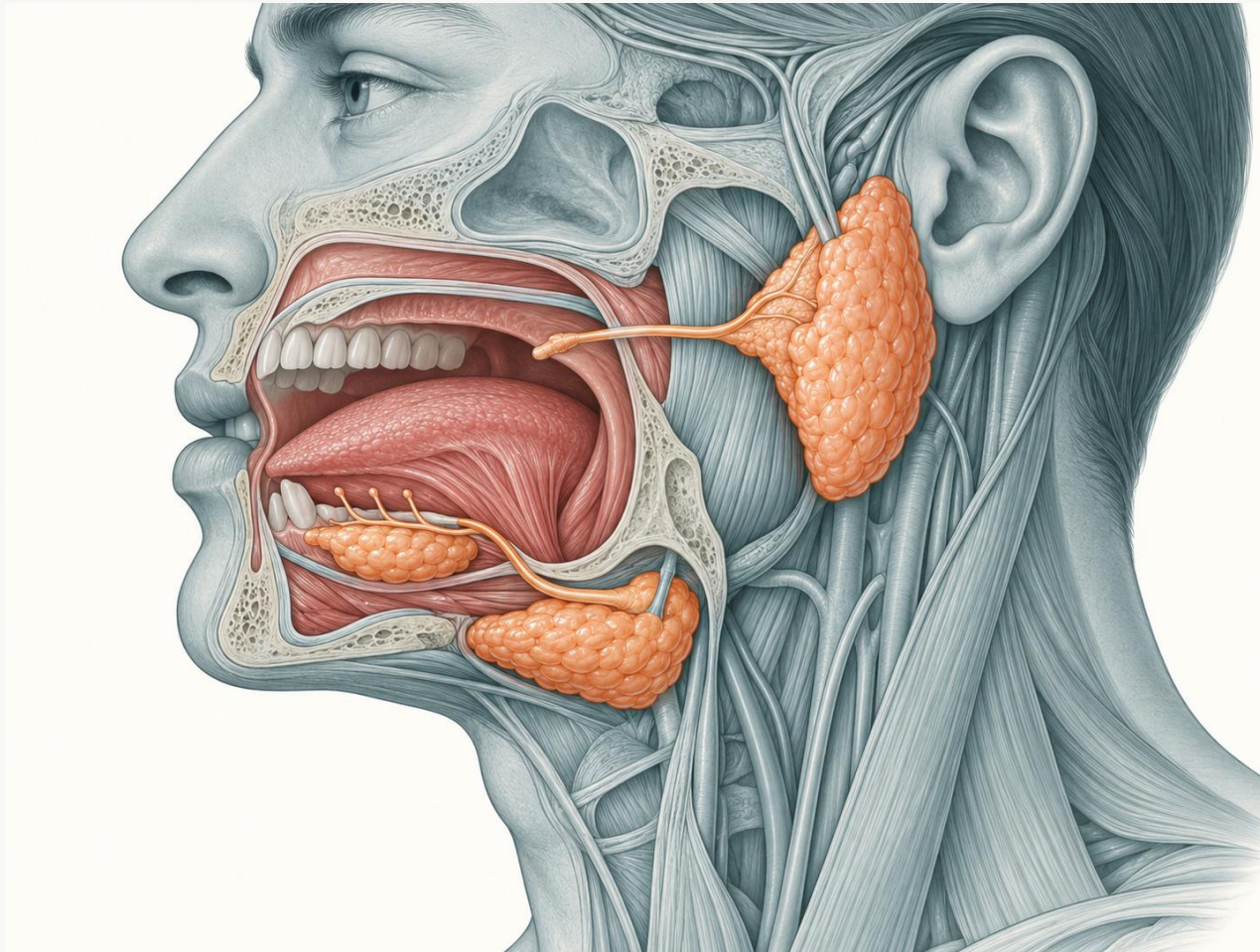


Learning outcomes

By the end of this lecture, you should be able to:

- 1 Explain how major salivary glands produce and modify saliva.
- 2 Relate salivary flow rate to electrolyte composition and function.
- 3 Compare parasympathetic and sympathetic control of salivation.
- 4 Describe oral, pharyngeal, and esophageal phases of swallowing.
- 5 Explain neural control, airway protection, and esophageal peristalsis.
- 6 **Apply physiology to xerostomia, aspiration, dysphagia, and achalasia.**

Major salivary glands: location predicts secretion



PAROTID Mostly serous • amylase-rich • contribution rises strongly during stimulation

SUBMANDIBULAR Mixed serous + mucous • major source of unstimulated saliva

SUBLINGUAL Mostly mucous • important for lubrication of the oral cavity

Minor glands continuously support mucosal lubrication.

What is in saliva?

WATER

~99% water
solvent + lubrication

ELECTROLYTES

Na^+ • Cl^- • K^+ • HCO_3^-
pH buffering + ionic balance

PROTEINS

α -amylase • mucins
IgA • lysozyme • lactoferrin

ORAL DEFENSE

Peroxidases + antimicrobial
peptides
protect teeth and mucosa

OTHER

Urea • uric acid • growth factors
+ small metabolites

Key idea: saliva is not “just water” — it is a protective, digestive, lubricating, buffering fluid.

Six functions of saliva

01

LUBRICATE

Mucins coat food and mucosa → easier speech, chewing, and swallowing.

02

DIGEST

Salivary α -amylase begins starch digestion; lingual lipase contributes to fat digestion.

03

TASTE

Dissolves molecules so they can reach taste receptors.

04

BUFFER

Bicarbonate helps neutralize acids; flow-dependent protection after meals.

05

PROTECT

Mechanical cleansing + IgA, lysozyme, lactoferrin, peroxidases.

06

TEETH

Ca^{2+} and phosphate support remineralization; reduced saliva increases caries risk.

Autonomic control of salivation

PARASYMPATHETIC

CN VII → submandibular + sublingual glands

CN IX → parotid gland

- Copious watery secretion
- Marked vasodilation + increased glandular blood flow
- Strongly supports digestion and oral lubrication

SYMPATHETIC

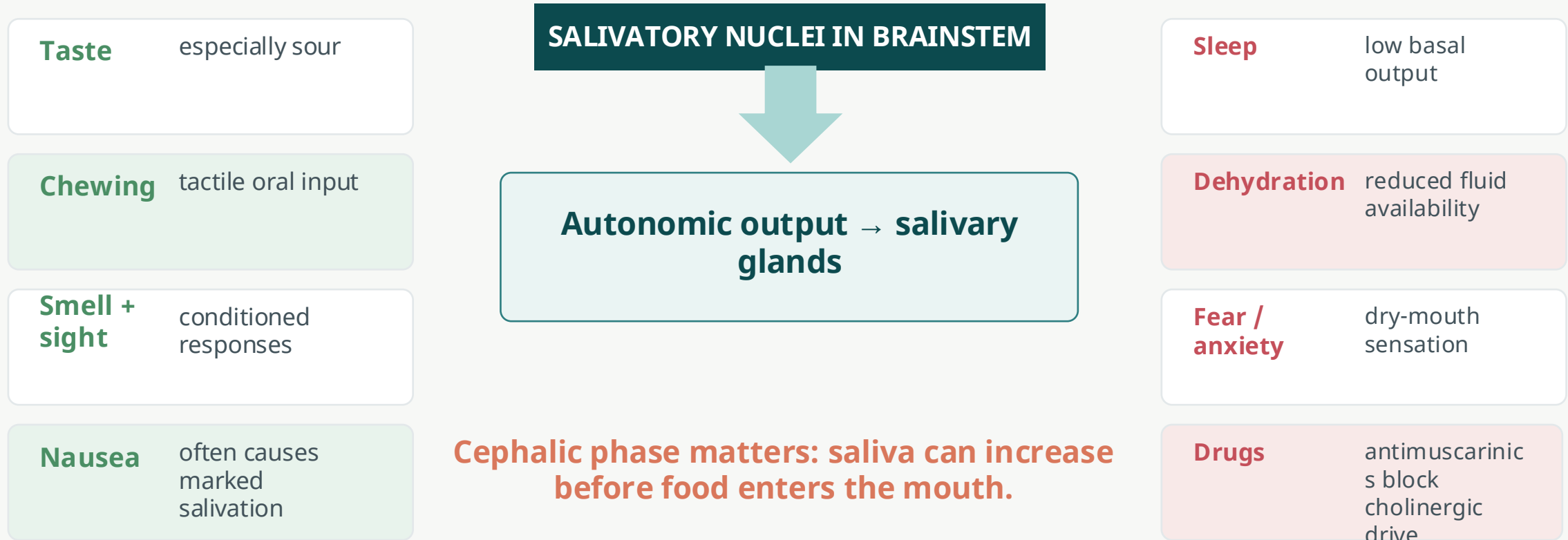
Thoracic sympathetic outflow

→ superior cervical ganglion → salivary glands

- Smaller-volume, protein-rich secretion
- Myoepithelial contraction contributes to expulsion
- Strong sympathetic states can feel “dry” despite some secretion

Both divisions can stimulate salivation — but the quantity and composition differ.

What increases — or decreases — salivation?



Clinical case: xerostomia after an antimuscarinic

CASE

A 62-year-old woman develops dry mouth after starting an antimuscarinic medication.

- Crackers feel difficult to swallow
- Taste is reduced
- She develops more dental caries over time

Which physiological functions of saliva explain these symptoms?

Mechanistic chain

↓ muscarinic stimulation



↓ watery salivary flow



↓ lubrication + solvent for taste



↓ **buffering / cleansing / remineralization**

Clinical case: pain and swelling during meals

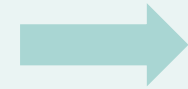
CASE

A 35-year-old man reports recurrent pain and swelling under the jaw, especially when eating.

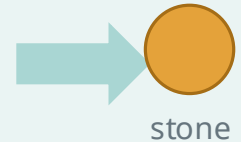
Why does eating make the pain worse?

Clue: the gland is being asked to increase flow against an obstructed duct.

Meal / taste stimulus



↑ salivary secretion



stone



Back-pressure → painful gland distension

Swallowing is a safety problem — not just a transport problem

Every swallow must solve three tasks in the correct sequence:

1

MOVE THE BOLUS

Propel food from oral cavity → pharynx → esophagus.

2

PROTECT THE AIRWAY

Pause breathing, close the laryngeal inlet, and prevent aspiration.

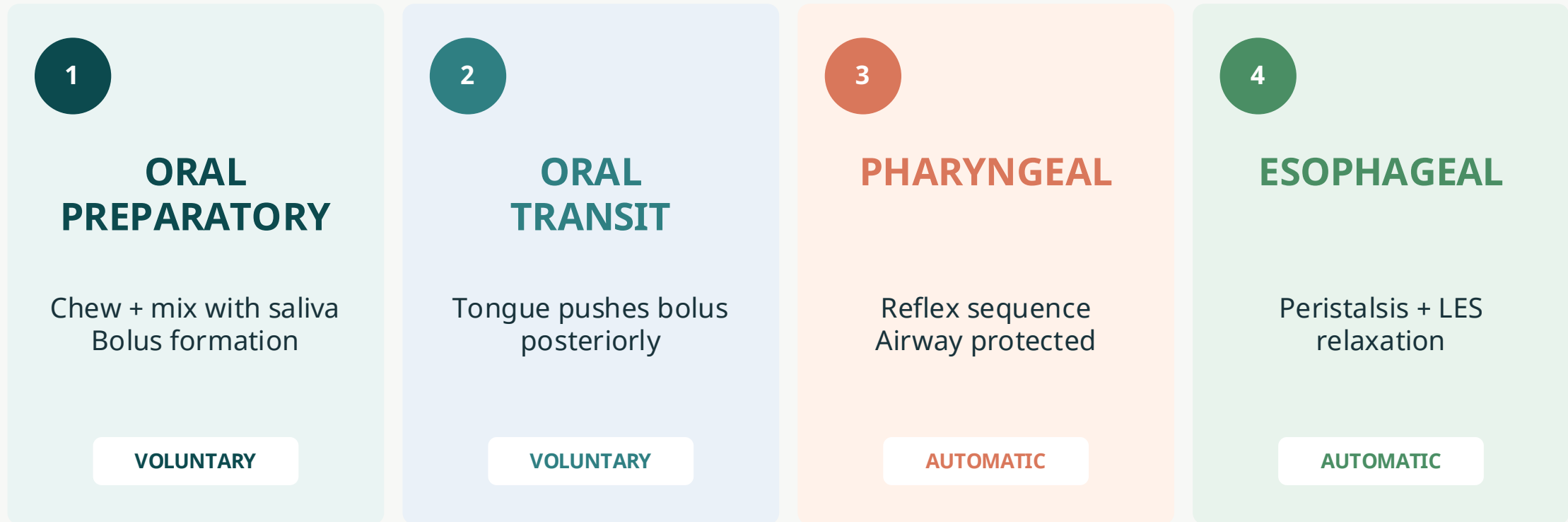
3

OPEN THE RIGHT SPHINCTERS

Coordinate upper esophageal sphincter opening and lower esophageal sphincter relaxation.

Timing is everything.

The phases of swallowing



The transition from voluntary to automatic control occurs as the bolus reaches the oropharynx.

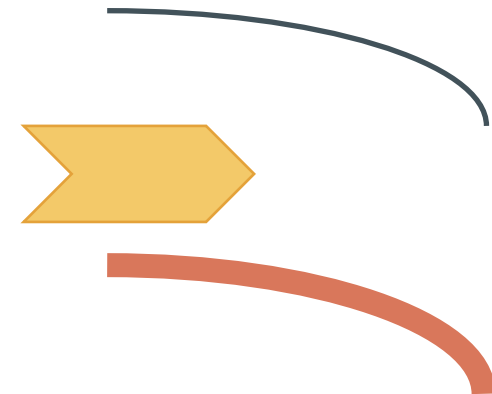
Oral phase: build the bolus, then launch it

ORAL PREPARATORY

- Chewing reduces particle size
- Saliva wets and binds particles
- Tongue shapes a cohesive bolus

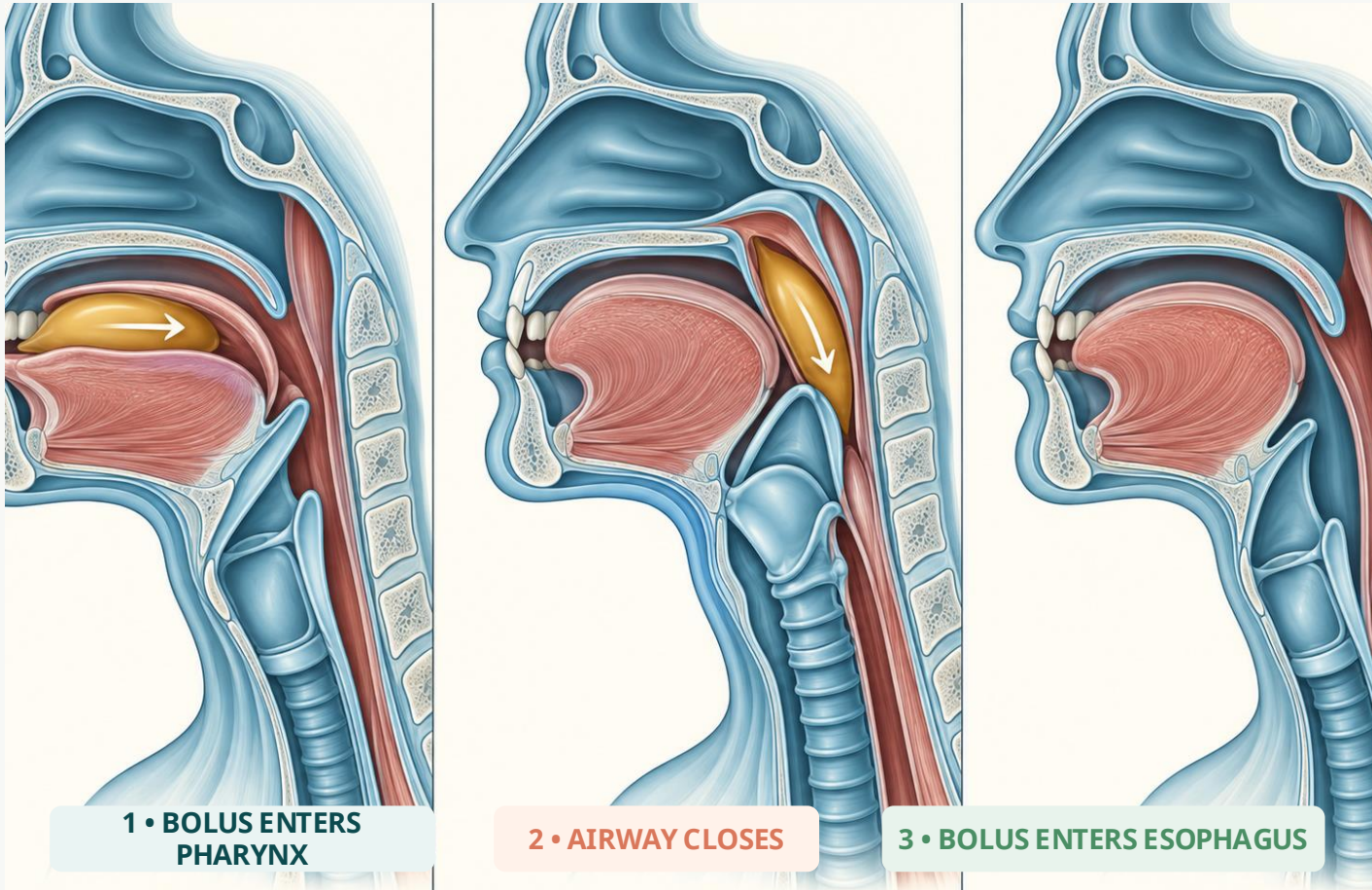
ORAL TRANSIT

- Tongue contacts hard palate
- Bolus is propelled posteriorly
- Sensory receptors trigger the pharyngeal sequence



Tongue pressure against palate
creates the propulsive force for oral transit

Pharyngeal phase: a rapid, programmed sequence

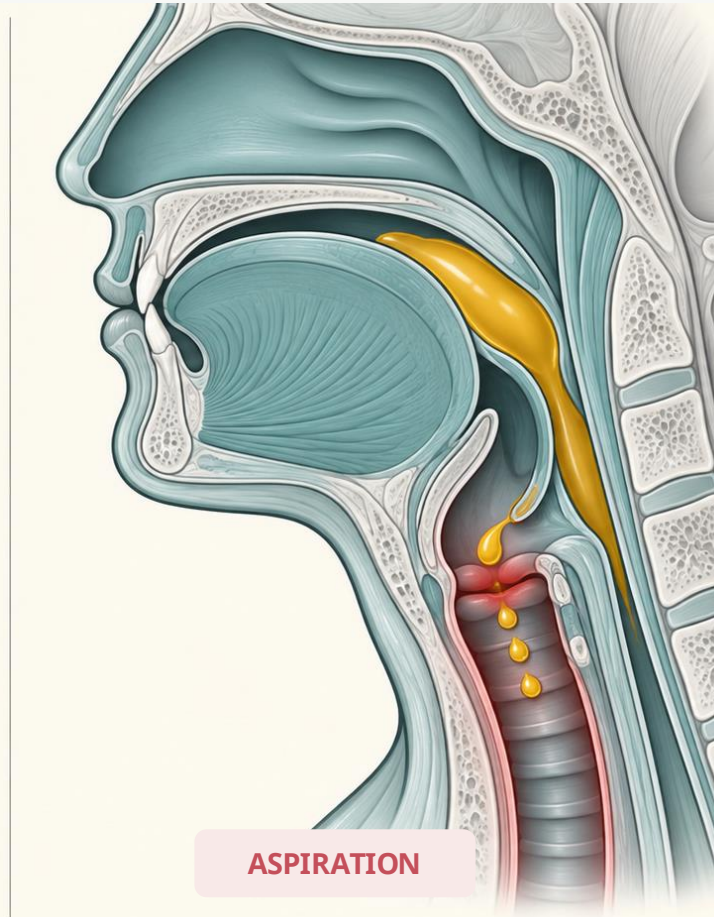
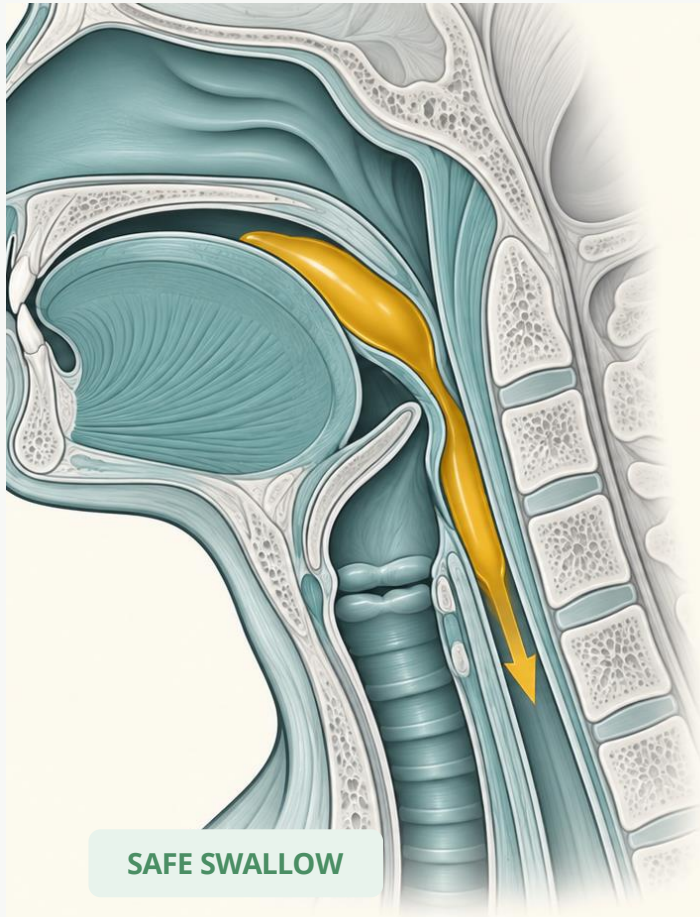


Core events

- Soft palate elevates → nasopharynx closes
- Larynx elevates and moves anteriorly
- Vocal folds adduct; epiglottis tilts
- Upper esophageal sphincter relaxes
- Pharyngeal constrictors propel bolus

**Breathing briefly pauses:
deglutition apnea.**

Airway protection: multiple barriers work together

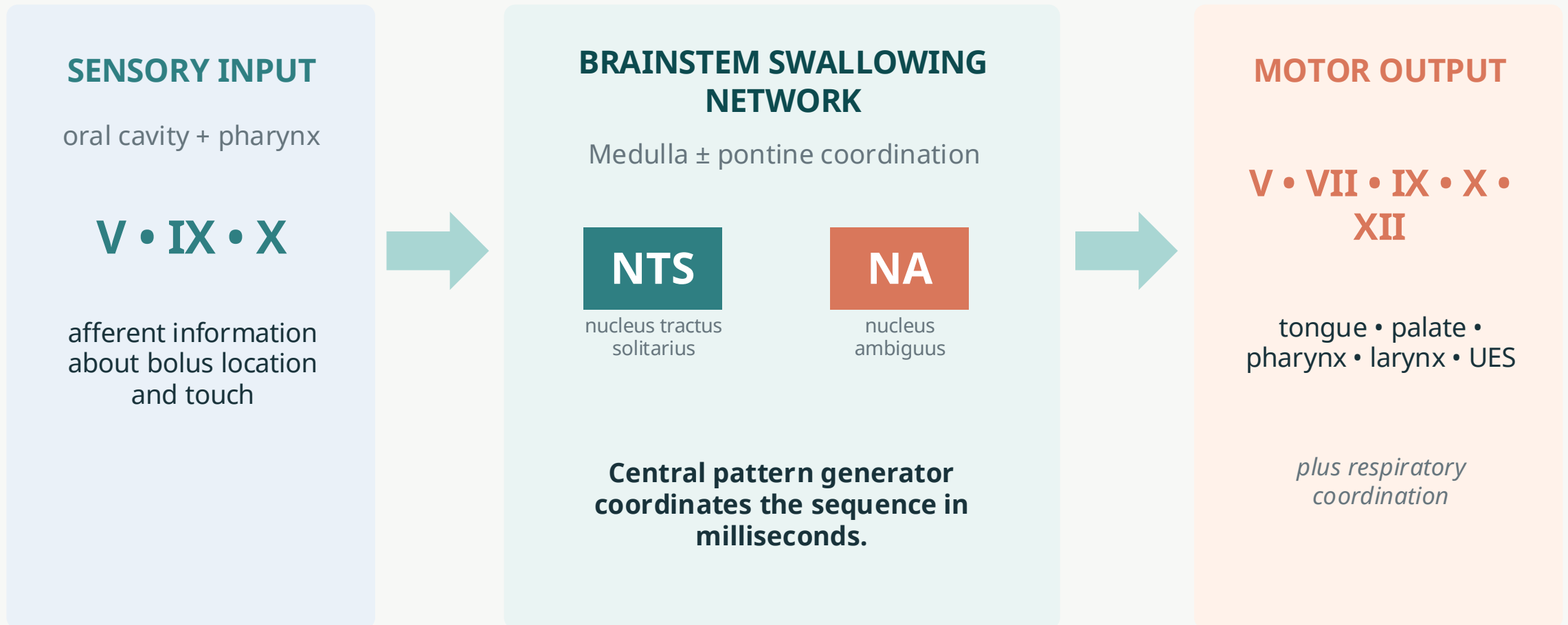


Protection is layered

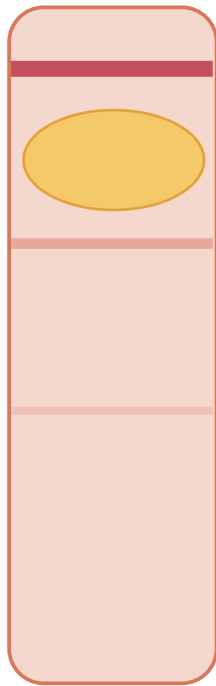
- True and false vocal folds close
- Arytenoids approximate
- Larynx elevates and moves forward
- Epiglottis redirects the bolus
- Cough reflex responds if material enters the airway

Epiglottis alone does NOT "seal" the airway.

Neural control of swallowing



Esophageal phase: peristalsis does the transport



Primary peristalsis

continuation of the
swallowing wave

Secondary peristalsis

triggered by esophageal
distension if material
remains

**Gravity helps — but it
is not required.**

LOWER ESOPHAGEAL SPHINCTER

Before bolus arrives

RELAXES

Deglutitive inhibition via vagal +
enteric inhibitory pathways (NO /
VIP)

Then tone returns after passage
of the bolus.

Clinical case: coughing immediately after water

CASE

A 73-year-old man after a stroke coughs immediately after drinking water. His voice becomes wet and gurgly.

Where is the problem?

- Difficulty is immediate
- Airway symptoms occur during/just after swallow
- Neurologic context

Most likely physiology

OROPHARYNGEAL / PHARYNGEAL DYSPHAGIA

Failed timing or strength of airway protection + pharyngeal propulsion → aspiration risk

RED FLAG: aspiration

Dysphagia: localize the physiology from the history

OROPHARYNGEAL

"I cannot start the swallow."

- Coughing/choking immediately
- Nasal regurgitation
- Wet voice
- Neurologic disease often present

**Problem zone: oral/pharyngeal control +
airway protection**

ESOPHAGEAL

"Food sticks after I swallow."

- Bolus sensation behind sternum
- Solids only → think mechanical obstruction
- Solids + liquids → think motility disorder
- Usually no immediate choking at initiation

**Problem zone: esophageal lumen, peristalsis,
or LES**

Clinical link: achalasia = failed inhibitory physiology

Normal

Swallow → **inhibitory neurons activate** → **LES relaxes ahead of bolus**

NO + VIP



LES OPENS

Achalasia

Loss of inhibitory myenteric neurons → **impaired LES relaxation + absent/abnormal peristalsis**

↓ inhibitory signaling



LES fails to relax

Physiology explains the symptom: solids and liquids both become difficult.

Take-home messages

- 1 Salivary acini make primary isotonic fluid; ducts modify it into mostly hypotonic saliva.
- 2 Flow rate matters: higher flow means less ductal modification.
- 3 Parasympathetic activity produces copious watery saliva; sympathetic activity changes quantity and protein content.
- 4 Swallowing moves from voluntary oral control to a rapid brainstem-programmed pharyngeal sequence.
- 5 Airway protection requires coordinated glottic closure, laryngeal movement, timing, and reflexes.
- 6 **Esophageal peristalsis plus LES relaxation completes transport — and explains clinical disorders when they fail.**

Understand the sequence. Then the clinical cases become predictable.

Recommended references

CORE TEXTS

Guyton and Hall Textbook of Medical Physiology

Chapters on gastrointestinal secretions, motility, and swallowing physiology.

Ganong's Review of Medical Physiology

Sections on salivary secretion, autonomic control, and gastrointestinal motility.

Costanzo Physiology

Concise review of GI secretions and motility for exam preparation.

Illustrations in this lecture were created specifically for teaching use.